

LaTeX Math Commands — Copy-Paste Reference

for Word / PowerPoint equation editors

from ASMC_Lyapunov_Stability_Proof.pdf

How to use: each block below is the raw LaTeX math content for one equation, in the order it appears in the proof. Copy any block and paste it into Word's or PowerPoint's equation editor (Insert → Equation → in LaTeX mode). The custom macros ($\backslash\mathbf{kmax}$, $\backslash\mathbf{bm}$, $\backslash\mathbf{abs}$, $\backslash\mathbf{norm}$, $\backslash\mathbf{diag}$, $\backslash\mathbf{sgn}$, $\backslash\mathbf{R}$) have been expanded to standard LaTeX so nothing extra is needed.

For align blocks (multiple equations stacked with $\&=$), each line is shown separately — you can paste each individually since Word/PowerPoint don't render align environments natively. The $\&$ alignment markers have been removed from the standalone lines for clarity.

Equation numbers match the proof PDF: “Eq. (15)” here is Eq. (15) there.

1 Implementation under analysis

Eq. (1) eq:ex

```
% Full align block:
e_x    &= (x_0 - x_d)\cos\psi + (y_0 - y_d)\sin\psi, \\
e_y    &= -(x_0 - x_d)\sin\psi + (y_0 - y_d)\cos\psi, \\
e_\psi &= \sin(\psi - \psi_d).
```

Individual lines (with $\&$ alignment removed):

```
e_x = (x_0 - x_d)\cos\psi + (y_0 - y_d)\sin\psi,
```

```
e_y = -(x_0 - x_d)\sin\psi + (y_0 - y_d)\cos\psi,
```

```
e_\psi = \sin(\psi - \psi_d).
```

Eq. (2) eq:edot_x

```
% Full align block:
\dot e_x    &= V_x - V_{x,d} + \dot\psi, e_y, \\
\dot e_y    &= V_y - V_{y,d} - \dot\psi, e_x, \\
\dot e_\psi &= \dot\psi - \omega_d.
```

Individual lines (with $\&$ alignment removed):

```
\dot e_x = V_x - V_{x,d} + \dot\psi, e_y,
```

```
\dot e_y = V_y - V_{y,d} - \dot\psi, e_x,
```

```
\dot e_\psi = \dot\psi - \omega_d.
```

Eq. (3) eq:lambda

```
\lambda_i(e_i) \;=\; \lambda_{i,\min} + (\lambda_{i,\max}-\lambda_{i,\min})\exp\{-\mu_i
\rightarrow e_i^2\bigr),
```

Eq. (4) eq:lambda_dot

```
\dot{\lambda_i} \;=\; -2\mu_i e_i,\dot{
↪ e_i},(\lambda_{i,\max}-\lambda_{i,\min})\,,\exp\!\bigl(-\mu_i e_i^2\bigr).
```

Eq. (5) eq:s

```
s_i \;=\; \dot{e_i} + \lambda_i(e_i)\,,e_i.
```

Eq. (6) eq:usw

```
u_{\mathrm{sw},i} \;=\; -K_i\,,\tanh(s_i/\varepsilon), \quad \varepsilon = 0.05.
```

Eq. (7) eq:Mxeq

```
% Full align block:
M_{x,\mathrm{eq}} \quad \&= R!\left[\,,(m_s + 4J_1/R^2)A_{x,\mathrm{eq}} - m_s\dot{\psi} V_y -
↪ m\,,a_Y\,,\alpha_{\mathrm{eq}} - m\,,a_X\,,\dot{\psi}^2 + \tfrac{4}{R^2}p_1V_x\right], \quad
M_{y,\mathrm{eq}} \quad \&= R!\left[\,,(m_s + 4J_1/R^2)A_{y,\mathrm{eq}} + m_s\dot{\psi} V_x +
↪ m\,,a_X\,,\alpha_{\mathrm{eq}} - m\,,a_Y\,,\dot{\psi}^2 + \tfrac{4}{R^2}p_1V_y + \tfrac{8}{R^2}p_2\right], \quad
↪ p_2\{(R-R_d)^2\}Bigr) V_y\right], \quad
M_{\psi,\mathrm{eq}} \quad \&= \bigl(I_s + 4(1+h)^2 J_1/R^2\bigr)\,,\alpha_{\mathrm{eq}} -
↪ m\,,a_Y(A_{x,\mathrm{eq}}-\dot{\psi} V_y) + m\,,a_X(A_{y,\mathrm{eq}}+\dot{\psi} V_x),
```

Individual lines (with & alignment removed):

```
M_{x,\mathrm{eq}} = R!\left[\,,(m_s + 4J_1/R^2)A_{x,\mathrm{eq}} - m_s\dot{\psi} V_y -
↪ m\,,a_Y\,,\alpha_{\mathrm{eq}} - m\,,a_X\,,\dot{\psi}^2 + \tfrac{4}{R^2}p_1V_x\right],
```

```
M_{y,\mathrm{eq}} = R!\left[\,,(m_s + 4J_1/R^2)A_{y,\mathrm{eq}} + m_s\dot{\psi} V_x +
↪ m\,,a_X\,,\alpha_{\mathrm{eq}} - m\,,a_Y\,,\dot{\psi}^2 + \tfrac{4}{R^2}p_1V_y + \tfrac{8}{R^2}p_2\right],
↪ p_2\{(R-R_d)^2\}Bigr) V_y\right],
```

```
M_{\psi,\mathrm{eq}} = \bigl(I_s + 4(1+h)^2 J_1/R^2\bigr)\,,\alpha_{\mathrm{eq}} -
↪ m\,,a_Y(A_{x,\mathrm{eq}}-\dot{\psi} V_y) + m\,,a_X(A_{y,\mathrm{eq}}+\dot{\psi} V_x),
```

Eq. (8) eq:Aieq

```
A_{i,\mathrm{eq}} \;=\; A_{i,d} - \dot{\lambda_i}, e_i - \lambda_i,\dot{e_i}, \quad
↪ i\in\{x,y,\psi\}.
```

Eq. (9) eq:wheelmap

```
\begin{pmatrix} M_1 \\\ M_2 \\\ M_3 \\\ M_4 \end{pmatrix}
\;=\;
\tfrac{1}{4}
\begin{pmatrix}
+1 & -1 & -L_{\psi} \\\
+1 & +1 & +L_{\psi} \\\
+1 & +1 & -L_{\psi} \\\
+1 & -1 & +L_{\psi}
\end{pmatrix}
\begin{pmatrix} M_x \\\ M_y \\\ M_{\psi} \end{pmatrix}.
```

Eq. (10) eq:sat

```
M_i^{\mathrm{sat}} \;=\; M_{\max}\,,\tanh\!\bigl(M_i / M_{\max}\bigr), \quad
↪ M_{\max}=10^{\text{N},m}.
```

Eq. (11) eq:Kdot

```
\dot K_i \;=\; \gamma_i\,\mathrm{bigl[\,s_i,\tanh(s_i/\varepsilon)\,\bigr]\,b(K_i) \;-\;
\rightarrow c_i\,(K_i/K_{\max})^3,
```

Eq. (12) eq:b

```
b(K) \;=\; \tfrac{1}{2} - \tfrac{1}{2} \tanh\!\bigl(K - (K_{\max}-2)\bigr).
```

2 Plant model on the sliding manifold

Eq. (13) eq:Mmat

```
\mathbf{M} \;=\;
\begin{pmatrix}
m_s & 0 & -m\,a_Y \\
0 & m_s & m\,a_X \\
-m\,a_Y & m\,a_X & I_s
\end{pmatrix},
```

Eq. (14) eq:plant

```
\mathbf{M}\,\dot{\mathbf{v}} \;=\; \mathbf{\tau}_{\mathrm{task}} + \mathbf{F}_{\mathrm{fric}}(\boldsymbol{\chi},\boldsymbol{\mu},\boldsymbol{\omega},\boldsymbol{\theta}) + \mathbf{C}(\mathbf{v}),
```

Eq. (15) eq:plant.reduced

```
\dot{\mathbf{v}} \;=\; \mathbf{A}_{\mathrm{eq}} + \mathbf{M}^{-1}\mathbf{\tau}_{\mathrm{sw}} +
\rightarrow \mathbf{M}^{-1}\mathbf{d}(t),
```

Eq. (16) eq:D

```
| \mathrm{bigl[\mathbf{M}^{-1}\mathbf{d}(t)\bigr]_i | \;\leq\; D_i, \quad \forall t \geq 0, \; i \in \{x,y,\psi\}.
```

3 Sliding-variable dynamics

Eq. (17) eq:sdot_raw

```
\dot s_i \;=\; \ddot e_i + \dot\lambda_i, \; e_i + \lambda_i, \; \dot e_i.
```

Eq. (18) eq:sdot_clean

```
% Full align block:
\dot s_i \;=\; \mathrm{Bigl}(\ddot{\mathbf{r}}_d - \dot\lambda, \; \mathbf{e} - \lambda, \; \dot{\mathbf{e}}) +
\rightarrow [\mathbf{M}^{-1}\mathbf{\tau}_{\mathrm{sw}}]_i + [\mathbf{M}^{-1}\mathbf{d}]_i - [\ddot{\mathbf{r}}_d]_i +
\rightarrow \dot\lambda, \; \mathbf{e} + \lambda, \; \dot{\mathbf{e}} \; \mathrm{nonumber} \\
&= [\mathbf{M}^{-1}\mathbf{\tau}_{\mathrm{sw}}]_i + [\mathbf{M}^{-1}\mathbf{d}]_i.
```

Individual lines (with & alignment removed):

```
\dot s_i = \mathrm{Bigl}(\ddot{\mathbf{r}}_d - \dot\lambda, \; \mathbf{e} - \lambda, \; \dot{\mathbf{e}}) +
\rightarrow [\mathbf{M}^{-1}\mathbf{\tau}_{\mathrm{sw}}]_i + [\mathbf{M}^{-1}\mathbf{d}]_i - [\ddot{\mathbf{r}}_d]_i +
\rightarrow \dot\lambda, \; \mathbf{e} + \lambda, \; \dot{\mathbf{e}} \; \mathrm{nonumber}
```

```
= [\mathbf{M}^{-1}\mathbf{\tau}_{\mathrm{sw}}]_i + [\mathbf{M}^{-1}\mathbf{d}]_i.
```

Eq. (19) eq:sx_full

$$\begin{aligned} \dot{s}_x &= -\mathbf{M}^{-1}_{11}K_x \tanh \sigma_x \\ &\quad - \underbrace{\mathbf{M}^{-1}_{12}}_{\text{cross-axis switching leaking into } x} K_y \tanh \sigma_y - \\ &\quad \mathbf{M}^{-1}_{13}K_\psi \tanh \sigma_\psi \\ &\quad + \mathbf{M}^{-1} \mathbf{d}_{\text{full}} \cdot \mathbf{x}. \end{aligned}$$

Eq. (20) eq:Def f

$$D_i^{\{\mathrm{eff}\}} \backslash ; = \backslash ; D_i^{\{\mathrm{raw}\}} \backslash ; + \backslash ; \sum_{j \neq i} | [\mathbf{M}^{-1}]_{ij} | \backslash , K_{\{\max\}},$$

Eq. (21) eq:beta

$$\beta_i \leq; [\mathbf{M}^{-1}]_{ii}, \quad \beta_x \leq 0.0282, \quad \beta_y \leq 0.0281, \quad \beta_\psi \leq 0.2274.$$

Eq. (22) eq:per_axis_canonical

$$\dot{s}_i \leq -\tilde{K}_i \tanh(s_i/\epsilon) + \tilde{d}_i, \quad |\tilde{d}_i| \leq D_i^{\mathrm{eff}},$$

4 Lyapunov function and its derivative

Eq. (23) eq:s_vec

```
% Full align block:
\dot{\bm s} \&= -\bm\beta\_,\bm K \odot \bm T(\bm s) + \tilde{\bm d}, \quad |\tilde{d}_i| \leq
\rightarrow D_i\sim{\mathrm{eff}}, \quad \backslash
\dot{\bm K} \&= \bm\Gamma_0,\bm\sigma(\bm s) \odot \mathrm{diag}(\bm b(\bm K)) - \bm c \odot
\rightarrow (\bm K/K_{\max})\sim{\mathrm{odot}} 3,
```

Individual lines (with & alignment removed):

$$\dot{\mathbf{b}}_s = -\mathbf{b}_s \beta, \mathbf{K} \odot \mathbf{T}(\mathbf{b}_s) + \tilde{\mathbf{d}}, \quad \text{quad } |\tilde{\mathbf{d}}_i| \leq \mathbf{D}_i^{\text{eff}},$$
$$\dot{\bm{K}} = \bm{\Gamma}_0, \bm{\sigma}(\bm{s}) \odot \mathrm{diag}(\bm{b}(\bm{K})) - \bm{c} \odot \dot{\bm{K}} \rightarrow (\bm{K}/K_{\max})^{\odot 3},$$

Eq. (24) eq:V

$$V_i(s_i, K_i) \leq \frac{s_i^2}{2} + \frac{\beta_i}{2\gamma_i} \log(K_i - K_i^* + 1)^2, \\ \quad \forall i \in \{x, y, \psi\}.$$

Eq. (25) eq:V_vec

$$V(\bm s, \bm K) \stackrel{\text{def}}{=} \frac{1}{s^{\text{top}}(\bm s)} + \frac{1}{(\bm K - \bm K^*)^{\text{top}}} \\ \rightarrow \frac{1}{\bm{\Gamma}^{-1}}, (\bm K - \bm K^*),$$

Eq. (26) eq:Vdot1

```
% Full align block:
\dot V_i \&= s_i\,,\dot s_i + \frac{\beta_i}{\gamma_i}(K_i - K_i^*)\,,\dot K_i \nonumber\\
&= s_i\bigl[-\beta_i K_i\,,\tanh(s_i/\varepsilon) + \tilde d_i\bigl] + \\
\rightarrow &\frac{\beta_i}{\gamma_i}(K_i - K_i^*)\Bigl[\frac{1}{\gamma_i},\sigma_i, b(K_i) - \\
\rightarrow &c_i\,,(K_i/K_{\max})^3\Bigr] \nonumber\\
&= -\beta_i K_i\,,\sigma_i + s_i\tilde d_i + \beta_i\,,(K_i - K_i^*)\,,\sigma_i,b(K_i) - \\
\rightarrow &\frac{\beta_i}{\gamma_i}(K_i - K_i^*)(K_i - K_i^*)(K_i/K_{\max})^3.
```

Individual lines (with & alignment removed):

$$\dot{V}_i = s_i, \dot{s}_i + \frac{\beta_i}{\gamma_i} (K_i - K_i^*) \dot{K}_i \quad \text{nonnumber}$$

$$\begin{aligned} &= s_i \bigl[-\beta_i K_i, \tanh(s_i/\varepsilon) + \tilde{d}_i \bigr] + \\ &\rightarrow \frac{\beta_i}{\gamma_i} (K_i - K_i^*) \bigl[\gamma_i, \sigma_i, b(K_i) - \\ &\rightarrow c_i, (K_i/K_{\max})^3 \bigr] \quad \text{nonnumber} \end{aligned}$$

$$\begin{aligned} &= -\beta_i K_i, \sigma_i + s_i \tilde{d}_i + \beta_i, (K_i - K_i^*) \dot{K}_i, \sigma_i, b(K_i) - \\ &\rightarrow \frac{\beta_i c_i}{\gamma_i} (K_i - K_i^*) (K_i/K_{\max})^3. \end{aligned}$$

Eq. (27) eq:Vdot2

$$\begin{aligned} \dot{V}_i &:= -\beta_i, \bigl[K_i, \sigma_i, (1 - b(K_i)) \dot{K}_i^*, \sigma_i, b(K_i) \bigr] \\ &\rightarrow \dot{s}_i, \tilde{d}_i - \frac{\beta_i c_i}{\gamma_i} (K_i - K_i^*) (K_i/K_{\max})^3. \end{aligned}$$

Eq. (28) eq:Vdot_vec

$$\begin{aligned} \dot{V} &:= -\sigma(s)^{\top} \beta, \bigl[K \odot (1 - b(K)) + \\ &\rightarrow K^* \odot b(K) \bigr] + s^{\top} \tilde{d} - (K - \\ &\rightarrow K^*)^{\top} \beta, \gamma_0^{-1} c \odot (K/K_{\max})^{\odot 3}. \end{aligned}$$

5 Region 1: adaptation active, $K_i \leq K_{\max}$

Eq. (29) eq:Vdot_r1

$$\begin{aligned} \dot{V}_i &:= -\beta_i, K_i^*, \sigma_i + s_i \tilde{d}_i - \frac{\beta_i c_i}{\gamma_i} (K_i - K_i^*) (K_i/K_{\max})^3. \end{aligned}$$

Eq. (30) eq:tanh_bound

$$\begin{aligned} \sigma_i &:= s_i, \tanh(s_i/\varepsilon) \geq |s_i|, \tanh(|s_i|/\varepsilon) \geq \\ &\rightarrow |s_i|, \tanh(1) \approx 0.7616, |s_i| \approx |s_i| \varepsilon. \end{aligned}$$

Eq. (31) eq:Vdot_r1_bound

$$\begin{aligned} \dot{V}_i &:= -\bigl[\beta_i, \tanh(1), K_i^* - D_i^{\mathrm{eff}} \bigr], |s_i| - \\ &\rightarrow \frac{\beta_i c_i}{\gamma_i} (K_i - K_i^*) (K_i/K_{\max})^3. \end{aligned}$$

Eq. (32) eq:cond_Kstar_pre

$$\begin{aligned} \beta_i, \tanh(1), K_i^* &> D_i^{\mathrm{eff}} \Leftrightarrow K_i^* > \\ &\rightarrow \frac{D_i^{\mathrm{eff}}}{\beta_i, \tanh(1)}. \end{aligned}$$

Eq. (33) eq:cond_Kstar

$$\begin{aligned} K_i^* &> \frac{D_i^{\mathrm{wrench}}}{\tanh(1)} \approx \\ &\rightarrow 1.313 D_i^{\mathrm{wrench}}. \end{aligned}$$

Eq. (34) eq:Vdot_compact

$$\begin{aligned} \dot{V}_i &:= -\alpha_{1,i}, |s_i| + \varepsilon_{0,i} \approx \alpha_{1,i} \equiv \\ &\rightarrow \beta_i \bigl[\tanh(1), K_i^* - D_i^{\mathrm{wrench}} \bigr] > 0. \end{aligned}$$

Eq. (35) eq:VKmax

$$\begin{aligned} V_{K,i} &:= V_{K,i}^{\max} \equiv \frac{\beta_i}{2\gamma_i}, \max \bigl[K_i^*, K_{\max} \\ &\rightarrow -K_i^* \bigr]^2. \end{aligned}$$

Eq. (36) eq:Vdot_sqrt

```
\dot V_i \;;\le\;; -\alpha_{1,i}\,\sqrt{V_i} \;;+\;; \epsilon_{0,i}, \quad V_i \ge 2
\rightarrow V_{K,i}^{\max}.
```

Eq. (37) eq:sqrt_rate

```
\frac{d}{dt}\sqrt{V_i} \;;\le\;; -\frac{\alpha_{1,i}^2}{2} + \frac{\epsilon_{0,i}^2}{2\sqrt{V_i}}.
```

Eq. (38) eq:sqrt_rate_clean

```
\frac{d}{dt}\sqrt{V_i} \;;\le\;; -\frac{\alpha_{1,i}^4}{4}.
```

Eq. (39) eq:sqrt_integrated

```
\sqrt{V_i(t)} \;;\le\;; \sqrt{V_i(0)} - \frac{\alpha_{1,i}}{4}t,
```

Eq. (40) eq:tstar

```
\;t_i^* \;;\le\;; \frac{4\bigl(\sqrt{V_i(0)} - \sqrt{V_i^{\mathrm{tail}}}\bigr)}{\alpha_{1,i}}.
```

Eq. (41) eq:Vdot_linear

```
\dot V_i \;;\le\;; -\kappa_i V_i + \tilde\epsilon_i, \quad
\kappa_i \equiv \frac{2\alpha_{1,i}}{M_{s,i}}, \quad
\tilde\epsilon_i \equiv \kappa_i V_{K,i}^{\max} + \epsilon_{0,i}.
```

Eq. (42) eq:gronwall

```
\;V_i(t) \;;\le\;; V_i(0)e^{-\kappa_i t} + \frac{\tilde\epsilon_i}{\kappa_i}\bigl(1 -
\rightarrow e^{-\kappa_i t}\bigr).
```

Eq. (43) eq:Vinf_lin

```
V_i^{\mathrm{lin},\infty} \;;=\;; \frac{\tilde\epsilon_i}{\kappa_i} \;;=\;; V_{K,i}^{\max} +
\rightarrow \frac{M_{s,i}}{\epsilon_{0,i}^2\alpha_{1,i}}.
```

Eq. (44) eq:bl

```
|s_i|_{\mathrm{ss}} \;;\lesssim\;; \frac{\varepsilon D_i^{\mathrm{eff}}}{\beta_i K_i} \;;=\;;
\rightarrow \frac{\varepsilon D_i^{\mathrm{wrench}}}{K_i}.
```

6 Region 2: gain saturating, $K_i \rightarrow K_{\max}$

Eq. (45) eq:Kdot_r2

```
\dot K_i \;;\approx\;; -c_i, (K_i/K_{\max})^3 \;;<\;; 0.
```

Eq. (46) eq:Vdot_r2

```
\dot V_i \;;\approx\;; -\beta_i K_i, \sigma_i \;;+\;; s_i \tilde d_i \;;-\;; \frac{\beta_i}{c_i \gamma_i} (K_i - K_i^*) (K_i/K_{\max})^3.
```

Eq. (47) eq:Vdot_r2_bound

```
\dot V_i \;;\le\;; -\bigl(\beta_i, \tanh(1)(K_{\max}-2) - D_i^{\mathrm{eff}}\bigr), |s_i| \;;-\;;
\rightarrow \frac{\beta_i c_i \gamma_i}{(K_{\max}-2 - K_i^*)}, ((K_{\max}-2)/K_{\max})^3.
```

7 Uniform ultimate boundedness

Eq. (48) unlabeled

$$K_{\max} > \frac{D_i^{\mathrm{wrench}} \tanh(1)}{1.313 D_i^{\mathrm{wrench}}},$$

8 From s to tracking error e

Eq. (49) eq:e_bound

$$\limsup_{t \rightarrow \infty} |e(t)| \leq \frac{\bar{s}}{\lambda_{\min}}.$$

Eq. (50) eq:e_bound_full

$$\limsup_{t \rightarrow \infty} |e(t)| \leq \frac{\varepsilon}{D(K, \lambda_{\min})}.$$

9 Stability bounds, summarized

Eq. (51) eq:sat_residual

$$\begin{aligned} \Delta^{\mathrm{sat}}_{\mathrm{wheel}} &= |M_i^{\mathrm{sw, wheel}}| - \\ &\rightarrow M_{\max} \tanh(\mathrm{bigl}(|M_i^{\mathrm{sw, wheel}}|/M_{\max}\bigr)) \approx 19 - 9.57 \\ &\rightarrow \approx 9.4 \text{ N}\cdot\mathrm{m}. \end{aligned}$$

10 Why the per-axis cubic coefficients matter

Eq. (52) eq:Keq

$$\begin{aligned} K_{\mathrm{eq}} &= \\ &\rightarrow K_{\max} \cdot \mathrm{Bigl}(\frac{\gamma}{\overline{s} \tanh(s/\varepsilon)})^c \mathrm{Bigr})^{1/3}. \end{aligned}$$